## 3.8.2 Magnetic field due to long straight conductor carrying current

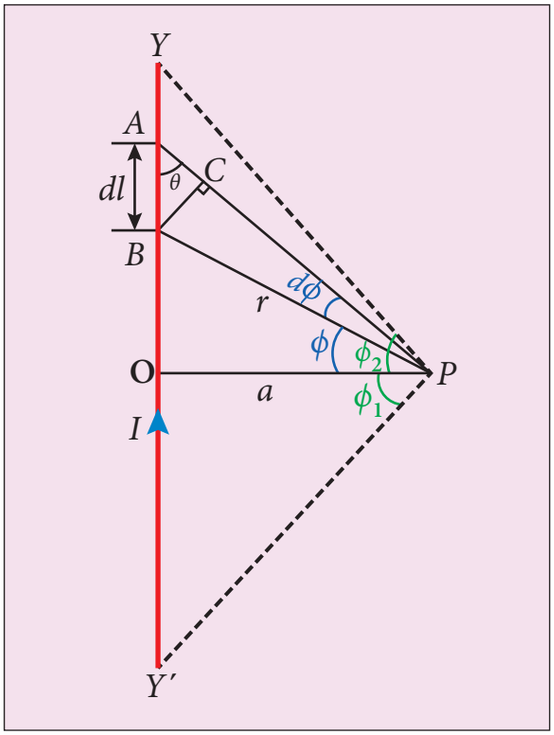



**Figure 3.32** Magnetic field due to a long straight current carrying conductor

Let YY′ be an infinitely long straight conductor and $I$ be the steady current through the conductor as shown in Figure 3.32. In order to calculate magnetic field at a point P which is at a distance $a$ from the wire, let us consider a small line element $dl$ (segment AB).

The magnetic field at a point P due to current element $Idl$ can be calculated from Biot-Savart's law, which is

$$d\vec{B} = \frac{\mu_0}{4\pi}\frac{Idl\sin\theta}{r^2}\hat{n}$$

where $\hat{n}$ is the unit vector which points into the page at P, $\theta$ is the angle between current element $Idl$ and line joining $dl$ and the point P. Let $r$ be the distance between line element at A to the point P.

To apply trigonometry, draw a perpendicular AC to the line BP as shown in Figure 3.32.

In triangle $\Delta ABC$, $\sin\theta = \frac{AC}{AB}$

$$\Rightarrow AC = AB\sin\theta$$

But $AB = dl \Rightarrow AC = dl\sin\theta$

Let $d\phi$ be the angle subtended between AP and BP

i.e., $\angle APB = \angle BPC = d\phi$

In a triangle $\Delta APC, \sin(d\phi) = \frac{AC}{AP}$

Since, $d\phi$ is very small, $\sin(d\phi) \simeq d\phi$

But $AP = r \Rightarrow AC = rd\phi$

$$\therefore AC = dl\sin\theta = rd\phi$$

$$\therefore d\vec{B} = \frac{\mu_0}{4\pi}\frac{I}{r^2}(rd\phi)\hat{n} = \frac{\mu_0}{4\pi}\frac{Id\phi}{r}\hat{n}$$

Let $\phi$ be the angle between BP and OP

In a $\Delta OPA$, $\cos\phi = \frac{OP}{BP} = \frac{a}{r}$

$$\Rightarrow r = \frac{a}{\cos\phi}$$

$$d\vec{B} = \frac{\mu_0}{4\pi}\frac{I}{a/\cos\phi}d\phi\hat{n}$$

$$\Rightarrow d\vec{B} = \frac{\mu_0 I}{4\pi a}\cos\phi d\phi\hat{n}$$

The total magnetic field at P due to the conductor YY′ is

$$\vec{B} = \int_{-\phi_1}^{\phi_2} d\vec{B} = \int_{-\phi 1}^{\phi_2}\frac{\mu_0 I}{4\pi a}\cos\phi d\phi\hat{n}$$

$$= \frac{\mu_0 I}{4\pi a}\left[\sin\phi\right]_{-\phi_1}^{\phi_2}\hat{n}$$



UNIT-3(XII-Physics_Vol-1).indd 156 14-02-2020 20:03:39